Grade 1 | Science | Planrs Academy

Colours in Black

The Science of Chromatography

Let's discover the rainbow hiding in black ink!

What Colour Is This?

Look at this black sketch pen. It looks like just one colour, right? But wait until you see what happens next!

Let's solve this mystery together!

Black hides a rainbow inside!

Black Sketch Pen ■

Black Is a Mixture

Black ink looks like one colour but is many mixed together!

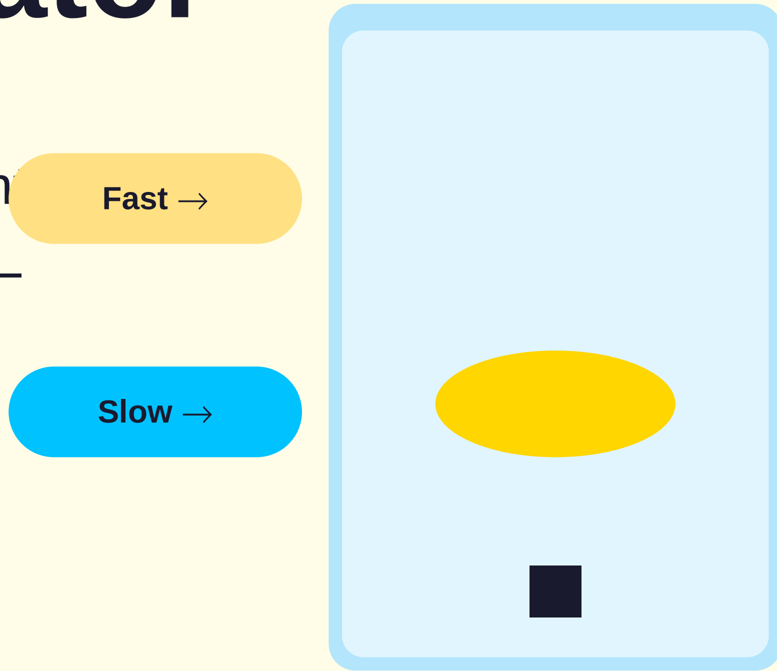
Like chaat masala — it looks brown but is really cumin, dry mango, black salt, and pepper all mixed up!

Orange
Pink
Cyan
Green
Purple
Yellow
Red

= BLACK!

Water as a Separator

Water can carry different colours at different speeds! Think of haldi (turmeric) in water — the yellow colour spreads out slowly and beautifully.



Haldi + Water

Paper as a Racetrack

Paper soaks water and gives colours a racetrack to run! Think of blotting paper in school or how roti soaks up dal — paper absorbs water the same way!

Let's Get Ready!

✏ Black Sketch Pen

Filter Paper

Water

Pencil / Toothpick

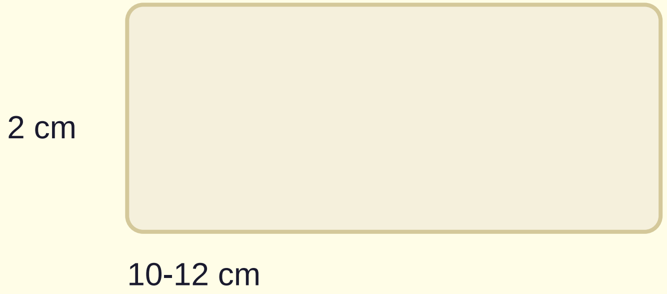
Small Katori

✂ Scissors

1 Step 1: Cut the Paper Strip

Cut a paper strip that is 2 cm wide and 10-12 cm long.

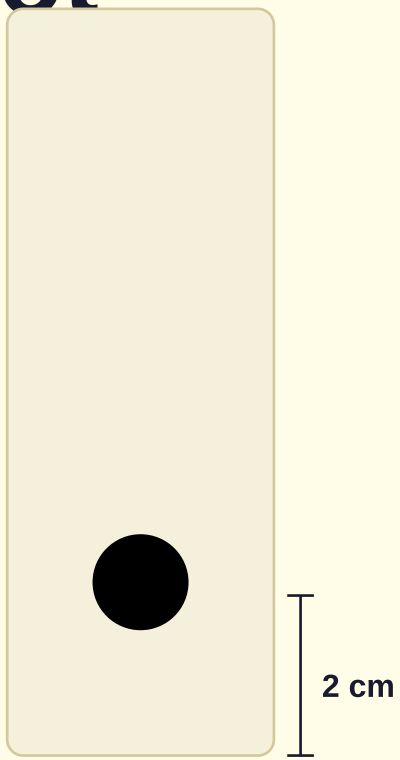
Ask an adult to help you with the scissors!



Step 2: Draw a Black Dot

Draw a thick black dot about 2 cm from the bottom of your paper strip. Use your ruler to measure!

Now wait 30 seconds for the ink to dry completely.

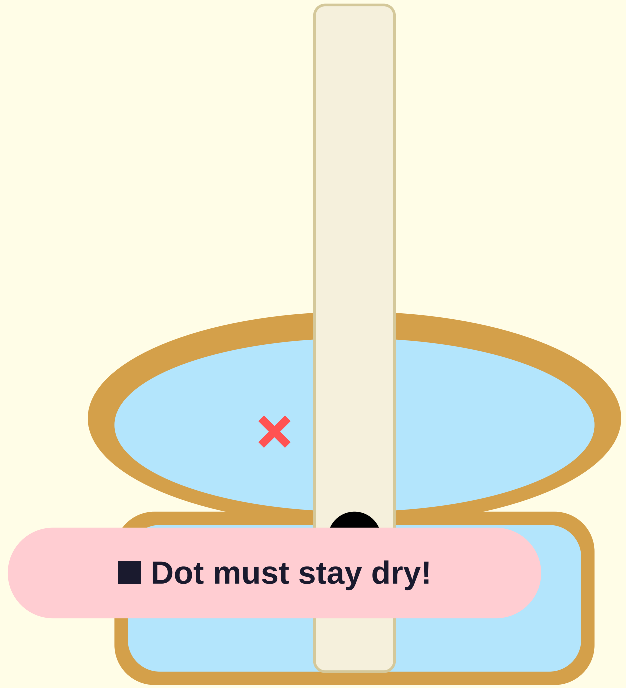


■■ Wait 30 seconds!

3 Step 3: Dip in Water

Dip the tip of your paper strip (below the dot) into the water in your katori.

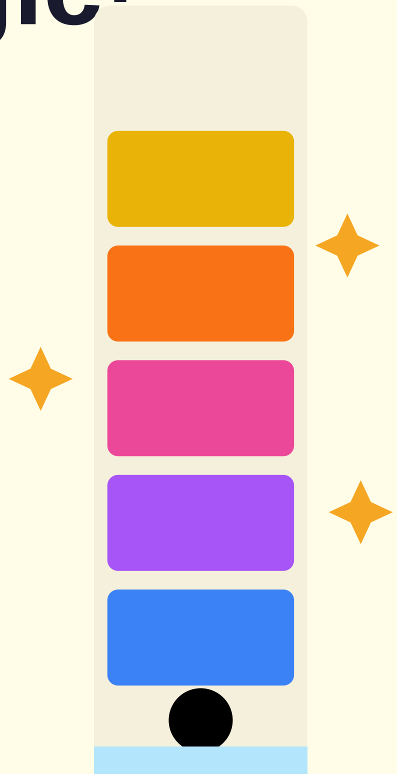
■■ Important: The black dot must NOT touch the water!
Only the very bottom of the paper should be in the water.



4 Step 4: Watch the Magic!

Hold the strip for 5-8 minutes. Watch carefully as water climbs up slowly and the colours start to separate like magic!

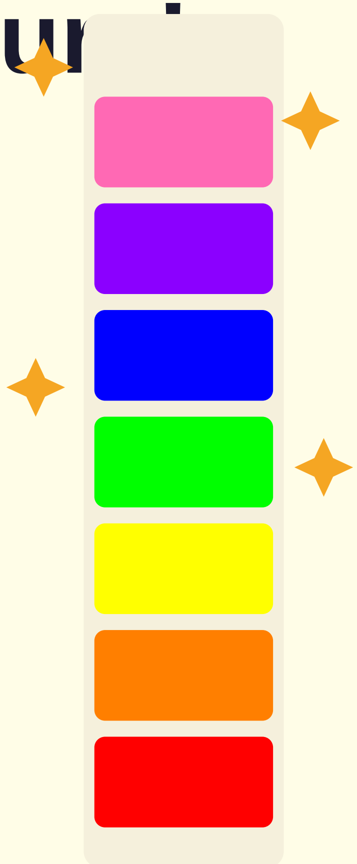
■ **Do NOT move or shake it!**



■■ **5-8 minutes**

5

Step 5: Look at the Colour



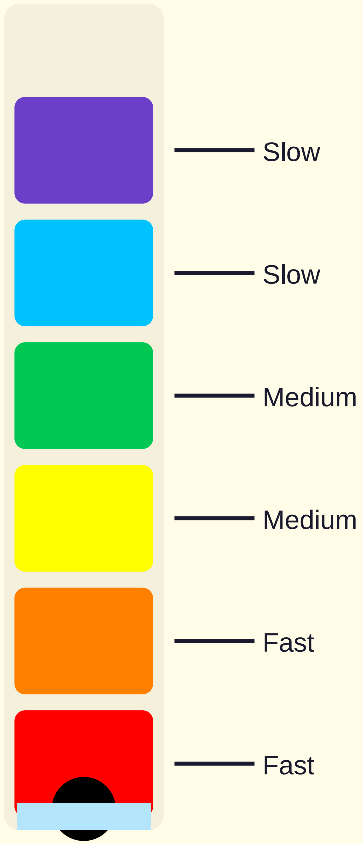
Remove your strip from the water and let it dry.

Now look closely at all the beautiful colours!

Draw or describe what you see on your strip.

Why Did That Happen?

- Paper is the racetrack for colours
- Water climbs up and carries colours
- Some colours run faster
- Some colours run slower



The Big Word:
CHROMATOGRAPHY

CHRO-MA-TO-GRAPHY

■ Clap it out!

Chroma = Colour

Graphy = Drawing or Writing

It means "Colour Writing!"

Scientists Use This!

- RBI checks if money is real
- Food scientists test mithai colours
- Police labs check handwriting
- Chromatography helps everyone!

Quiz Question 1

Black ink is made of ____________

Let's test what you learned about the colours hiding inside black ink!

A) Only one colour

B) Many colours mixed together

Question 1 of 3

Quiz Question 2

In our experiment, what helped separate the colours?

A) Salt

B) Water

Question 2 of 3

Quiz Question 3

The name of separating colours using paper and water is called ________

A) Photography

B) Chromatography

Question 3 of 3

Planrs Academy

Junior Chromatographer!

"Ink looks black — but that's a trick! Science shows what colours STICK!
Water climbs up, colours spread wide — A rainbow was hiding right inside!"